

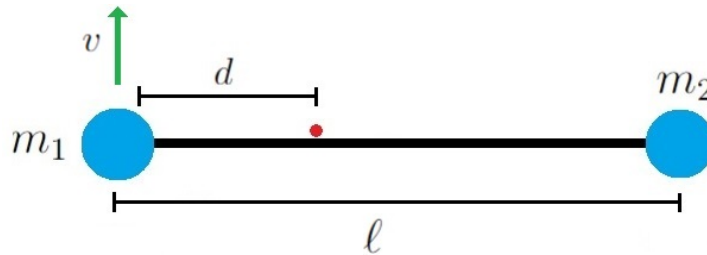
Quirky Problems from the Classical Mechanics and Thermodynamics Course 2020/21

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1 Kinematics and Dynamics

Problem 1.1 (Bicycle Wheel). Let us imagine taking a photograph of a bicycle wheel rolling without slipping. The wheel can be modeled as a circle connected to its center by a large number of spokes. Since, when taking a photo, the camera shutter remains open for a small but nonzero time interval δt , most of the photo will appear blurred because the wheel's parts move during the exposure time. Yet, there are some points where the photo appears perfectly sharp. Which points are these?

Problem 1.2 (Motion of Two Masses on a Table). Two point-like bodies of masses m_1 and m_2 are connected by a massless, inextensible string of length ℓ . They are placed on a perfectly smooth horizontal table. The following figure shows the top view of the configuration (the sizes of the bodies are exaggerated).



The red point in the figure represents a nail fixed to the table, initially at a distance d from the left-hand body. At the initial instant, the string is perfectly taut, and an impulse is given to the left body, which acquires a velocity v perpendicular to the string.

1. Taking the nail as the origin, find the trajectories of both bodies from the initial instant until the moment when the body of mass m_2 touches or passes over the nail. How much time elapses between these two instants?
2. Over the same time interval, what is the minimum distance between the two bodies?
3. After the second body passes the nail, what kind of motion do the two bodies perform? Determine the quantities that characterize it—magnitude, direction, and sense.

Problem 1.3 (Atwood Machine with Massive String). An Atwood machine consists of a pulley of negligible size and two blocks of masses m_1 and m_2 , connected by a homogeneous string of length $2l$ and mass M . Initially, the blocks are at the same height, and then they are released. Find the function $y(t)$ that describes the system's time evolution up to the moment when the lighter block hits the pulley.

Problem 1.4 (Non-Uniform Rough Plane). A plane is composed of several materials arranged in wedge-shaped sectors as shown in Fig. 1. The N sectors all touch at the origin and have different angular openings θ_i , such that $\sum_{i=1}^N \theta_i = 2\pi$. Each sector is characterized by a friction coefficient μ_i .

A homogeneous disk of radius R and mass M is set into rotation about its axis with angular velocity $\vec{\omega} = \omega \hat{z}$ and then placed on the plane, with its center coinciding with the origin.

1. What is the disk's acceleration vector immediately after it touches the surface?
2. What is its initial angular acceleration?
3. What happens after the initial instant?

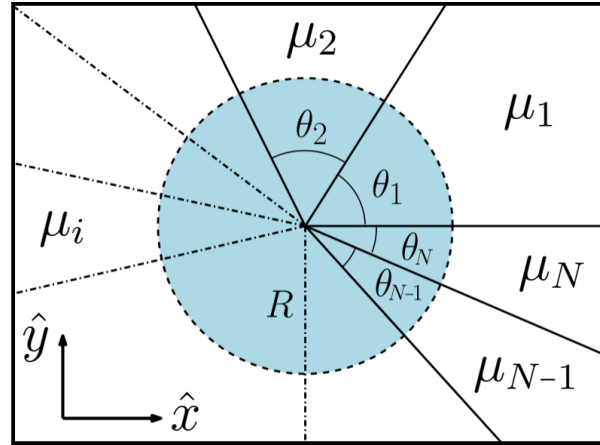


Figure 1: Disk placed on a rough plane.

2 Gravitation and Central Forces

Problem 2.1 (L4 and L5). A planet of mass m orbits a star of mass $M \gg m$ in a circular orbit. Find the positions of the Lagrangian points L4 and L5.

Problem 2.2 (Circular Orbit). A particle of mass m is subjected to a central force such that its orbit is described by the polar equation $r(\theta) = 2R \cos \theta$. Find the orbital period of this orbit as a function of **only** m , R , and the force parameter.

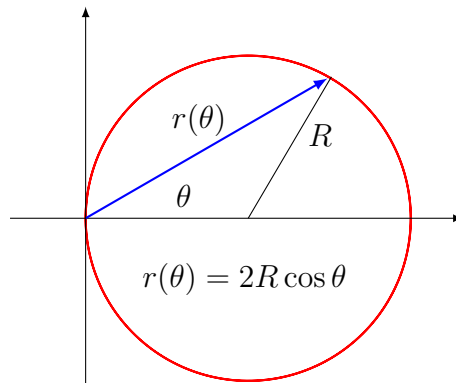


Figure 2: Circular orbit.

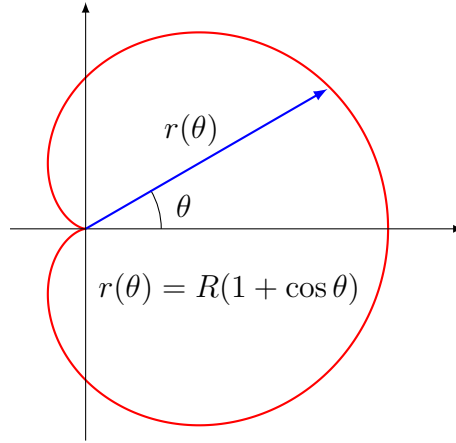


Figure 3: Cardioidal orbit.

Problem 2.3 (Cardioid). A particle of mass m is subjected to a central force such that its orbit is described by the polar equation $r(\theta) = R(1 + \cos \theta)$. Find the orbital period as a function of **only** m , R , and the force parameter.

Problem 2.4 (LRL Vector). Find the equation of the orbits starting from the Laplace–Runge–Lenz vector

$$\vec{A} = \vec{p} \times \vec{L} - m\alpha \hat{r}.$$

Problem 2.5 (Ballistic Missile). A ballistic missile is launched from a point A on Earth's surface with an initial velocity $\vec{v}_0$, whose magnitude is lower than the escape velocity, and at an angle α from the horizontal. After following an elliptical arc, the missile impacts the surface at some point B , whose position depends on the launch conditions. The situation is illustrated in Fig. 4.

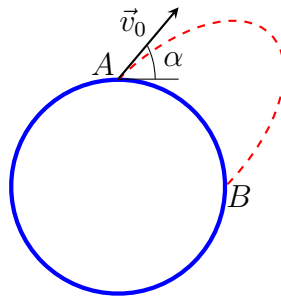


Figure 4: Example of projectile motion within Earth's gravitational field.

1. Find the orbital parameters as functions of v_0 and α .
2. Find the flight time and the distance between the launch and impact points on the surface. How far is the highest point of the trajectory from the Earth's surface?
3. Determine the initial velocity v_0 required for a missile launched at a given angle α to impact at the antipode, as shown in Fig. 5.

Problem 2.6 (Asteroid in the Solar System). An asteroid of mass m moves along a parabolic trajectory around the Sun, in the same plane as Earth's orbit, which is assumed circular with radius R . Let p be the asteroid's minimum distance from the Sun. Neglect the gravitational attraction between the asteroid and Earth.

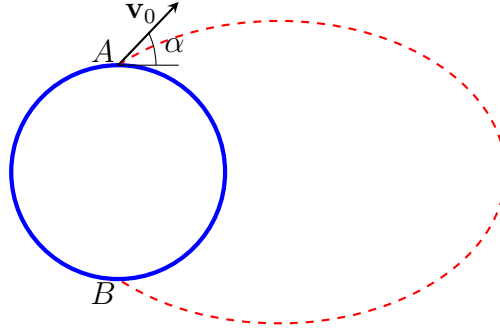


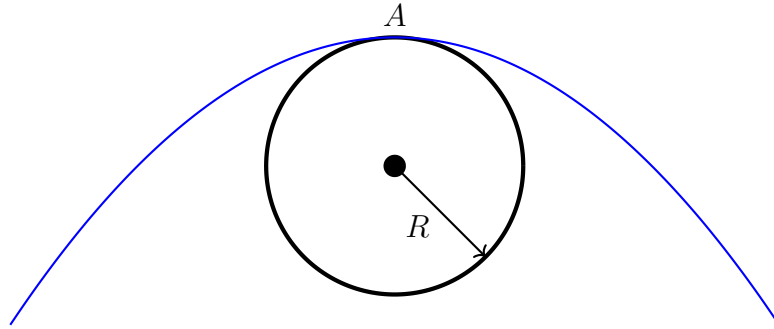
Figure 5: Projectile motion from a point toward the antipode.

1. What is the asteroid's angular momentum?
2. For $p < R$, what is the time T the asteroid spends inside Earth's orbit?
3. What is the maximum possible value of T ?

If p is smaller than the Sun's radius, the asteroid will fall into it, increasing the Sun's mass by a fraction $\alpha \equiv m/M_S$.

4. How do Earth's orbital parameters vary as a function of α ?

If $p = R$, the parabola is tangent to the circle at one point, as shown in the following figure.



Suppose the asteroid collides inelastically with Earth at point A , embedding itself into it.

5. How do Earth's orbital parameters change?
6. How much heat is released as a result of the collision?

3 Rigid Body

Problem 3.1 (Rigid Body in Orbit). A satellite orbits Earth on a circular trajectory of radius R . Compute the torque about its center of mass, given the inertia tensor $\bar{I}$ of the satellite. Initially, the three principal axes are oriented such that one is radial, one tangential, and the third perpendicular to the orbital plane. Show that this orientation remains fixed in the co-rotating frame attached to the satellite. Calculate the period of small oscillations within the orbital plane.

Problem 3.2 (Asymmetric Top). A rigid body with principal moments of inertia I_1 , I_2 , and I_3 (with $I_1 < I_2 < I_3$) moves freely without external forces, with total angular momentum $\vec{J} \equiv J\hat{z}$. The body's principal axes are denoted $\hat{n}_1$, $\hat{n}_2$, $\hat{n}_3$, while the inertial reference frame axes are $\hat{x}$, $\hat{y}$, and $\hat{z}$. At $t = 0$, the body's $\hat{n}_3$ axis makes a small angle $\theta_0 \ll 1$ with $\hat{z}$, and the transverse component of the angular velocity ω_T^0 is much smaller than the parallel component ω_3^0 .

1. Determine the time dependence of the angular momentum components $\vec{J}$ along the three principal axes, choosing the time origin appropriately.
2. Write the time laws of the Euler angles θ , ψ , and ϕ .
3. Determine how the unit vector $\hat{n}_3$ moves with time in the inertial frame, expressing it in terms of its coordinates (n_{3x}, n_{3y}, n_{3z}) .
4. Now consider an asymmetric top with principal moments $I_1 = A - \Delta$, $I_2 = A + \Delta$, $I_3 = 2A$ (with $0 < \Delta < A$) and generic initial conditions. Show that the transverse component ω_T of the angular velocity perpendicular to $\hat{n}_3$ remains constant in time, and write the equation of motion for ω_3 accordingly.

4 Calculus of Variations and Lagrangians

Problem 4.1 (Motion of a Charge). Using the Lagrangian formalism, study the motion of a particle with charge q subjected to a central force $\vec{F} = f(r)\hat{r}$ and a uniform magnetic field. Find a cyclic coordinate and the corresponding conserved conjugate momentum. Provide a physical interpretation of this conservation law.

Problem 4.2 (Chain of Springs). A system consists of $N + 1$ springs of rest length L connecting N point masses of mass m , forming a linear chain whose ends are fixed a distance $(N + 1)L$ apart. Write the Lagrangian of the system in terms of N generalized coordinates describing the displacements from equilibrium, and determine the spectrum of the N normal mode frequencies.

Problem 4.3 (Maximal Gravitational Field). We have a piece of plasticine of total mass m and uniform density ρ , and we wish to shape it to produce the maximum possible gravitational field at a given point in space. What shape should it have? What is the value of the maximal field obtained?

Problem 4.4 (Maximal Gravitational Field in 2D). Repeat the calculations of the previous problem for a 2D case, where the gravitational field has a different functional form.

Problem 4.5 (Snell's Law for Spherically Symmetric Media). Determine how Snell's law is expressed in the case of spherically symmetric media, characterized by a refractive index $n = n(r)$.

Problem 4.6 (Light Ray Trajectories). In a desert, the refractive index depends on the altitude according to the law

$$n(z) = n_0 \sqrt{1 + z/h},$$

where $n_0 > 1$ and h is a characteristic length. Find the function $z(x)$ describing the trajectory of a light ray that starts at point $A = (0, h)$ and reaches an observer at $B = (h, h)$.

Problem 4.7 (Superior and Inferior Mirages). Find the conditions on the air's refractive index $n = n(z)$ under which inferior and superior mirages can occur.

5 Thermodynamics

Problem 5.1 (Leaking Container). Inside a container of volume V , whose walls are perfect thermal insulators, there is a monoatomic ideal gas of molecular mass m , density ρ_0 , and temperature T_0 . The container is placed in a perfect vacuum. A small hole of diameter d , smaller than the mean free path of the gas molecules, is opened on one wall.

- Assuming that the gas temperature inside the container does not change over time, what is the law describing how the gas density $\rho(t)$ varies?
- The previous assumption is incorrect: the temperature changes because the faster molecules escape before the slower ones. What is the correct expression for $\rho(t)$?

Problem 5.2 (Two Compartments). A thermally insulated container is divided into two compartments, each containing the same number of monoatomic gas molecules. Initially, both compartments have the same temperature T_0 , pressure P_0 , and volume. As shown in Fig. 6, two pistons are connected to walls 1 and 3 (which do not conduct heat), while wall 2 is fixed and is a perfect thermal conductor. The initial distance between walls 1 and 2 is L , the same as between walls 2 and 3. At a certain moment, both pistons are moved simultaneously, displacing walls 1 and 3 by $L/2$ to the right. The process is slow enough for the two compartments to remain in thermal equilibrium. What is the final temperature of the gas once the process is complete?

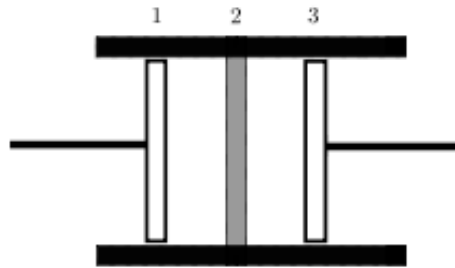


Figure 6: Compartments and pistons.

Problem 5.3 (Equation of State). Determine the equation of state $f(V, P, T) = 0$ of a classical gas of N particles, using *only* the following information: at constant temperature, its internal energy does not depend on volume, and its enthalpy does not depend on pressure.